

ANIRUDH BHATTACHARYA

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Education

Jaypee University of Engineering and Technology

2019 - 2023

Bachelor of Technology in Electronics and Communications Engineering | CGPA: 7.4

Guna, Madhya Pradesh

Relevant Coursework

- VLSI Design • Digital Hardware Design • Digital Circuit Design • Analogue Electronics
- Semiconductor Devices • Static Timing Analysis • Physical Design Flow • CMOS Fundamentals

Experience

Synopsys Inc

Apr 2025 - Present

Physical Design Application Engineer

Hyderabad, Telangana

- Support RTL-to-GDS Physical Design on customer ASIC / SoC blocks: floorplanning, placement, CTS, routing, timing closure, IR/EM, and PV signoff on Fusion Compiler, ICC2, PrimeTime, Calibre.
- Debug P&R, STA, congestion, and signoff failures; automate reusable checks in Tcl, Perl, Python. Work with AE, support, and design teams to root-cause QoR, tool, and methodology issues.
- Packaged Qualcomm P&R debug as GSAE testcase `gsaetc_fc_command_trace`; used as a production validation / regression check by multiple PD teams (Excellence Award, Mar 2026).
- Close 100+ tickets on FinFET-class nodes (TSMC N5A): timing, congestion, DRC/LVS, IR/EM, flow. Used DSO.ai for PPA exploration on a PVT Sensor IP as ML-assisted Physical Design.

Synopsys Inc

Apr 2024 - Mar 2025

Physical Design Intern

Hyderabad / Bhubaneswar

- Completed intensive RTL-to-GDSII implementation and signoff training: floorplan, P&R, CTS, STA, PDN / power planning, physical verification (DRC/LVS) on Fusion Compiler.
- Implemented DSO.ai-driven optimization for a PVT Sensor IP, improving timing, power, and block-level PPA/QoR with AI-assisted Physical Design methodology.
- Cleared all theoretical and practical assessments and converted to full-time Physical Design AE.

VLSI Expert Private Limited

May 2022 - Feb 2023

VLSI Trainee - Physical Design

Noida, Uttar Pradesh

- Implemented RISC-V picorv32a RTL-to-GDS: floorplan, PDN, CTS, P&R, STA, and signoff setup in ICC2 and PrimeTime, reading Verilog RTL to close timing and PPA.
- Developed SDC constraints and max/min parasitic corners for MCMM STA and signoff readiness.

Projects

DSO.ai Optimization Flow for PVT Sensor IP | Fusion Compiler, DSO.ai

Nov 2024 - Jan 2025

- AI-assisted PD optimization focused on timing closure, power, and PPA/QoR for the PVT Sensor IP. Used Synopsys implementation methodology on a hard-IP class block.

RISC-V picorv32a Floorplan and Signoff | ICC2, PrimeTime, MCMM

Sep 2022 - Dec 2022

- 90% core utilization with PDN / power planning, CTS, routing, STA, and physical-verification signoff. Performed stage-wise QoR and timing closure targeting RTL-to-GDS completion.

MCMM, Constraints & Parasitic Setup | Design Compiler, ICC2, PrimeTime

Nov 2022 - Dec 2022

- Configured RTL-to-GDS environment with SDC, MCMM scenarios, and max/min parasitics. Scripted the flow in Tcl / Python.

Technical Skills

Languages: Tcl, Python, Perl, Make, Verilog, C/C++

CAD / EDA Tools: Fusion Compiler, ICC2 / IC Compiler II, PrimeTime, DSO.ai, Design Compiler, Calibre

Physical Design: RTL-to-GDS, Floorplan, Placement, CTS, Routing, STA, Timing closure, PDN, IR/EM, PV (DRC/LVS/Antenna), PPA, MCMM, SDC, Low power, Block / hierarchical ASIC-SoC, Linux automation

Leadership / Extracurricular

IETE Students Forum (Electronics Club)

Sep 2019 - Aug 2021

Volunteer

Jaypee University of Engineering and Technology

VR AR MR Club

Sep 2019 - Oct 2020

Coordinator

Jaypee University of Engineering and Technology